

Chromium Diffusivity in 316 Alloys

Evidence and uncertainty at 700 °C

Main finding

The inspected evidence supports a provisional lattice-tracer range of approximately 3×10^{-20} – 6×10^{-19} m²/s at 973.15 K. This is a between-study sensitivity envelope, not a statistical confidence interval. Simulated, grain-boundary and corrosion-inferred coefficients require separate interpretation.

What parameter are we varying?

A tracer diffusivity describes the random migration of a tagged species in a specified material state. It differs from the chemical diffusion matrix that relates composition gradients to coupled fluxes. Our baseline is the Cr lattice-tracer fit for Fe–17 wt% Cr–12 wt% Ni reported in the ORNL/Perkins work, a ternary surrogate rather than a complete commercial 316 composition.[1]

$$D_{\text{Cr},0}^* = 1.3 \times 10^{-5} \exp \left[-\frac{264010.4}{RT} \right] = 8.7749 \times 10^{-20} \text{ m}^2/\text{s}. \quad (1)$$

Equation meaning. The prefactor is in m²/s, the activation energy in J/mol, and $R = 8.31446 \text{ J}/(\text{mol K})$. The original 63100 cal/mol is converted using 4.184 J/cal.

Experimental lattice fits

Source	D_0 (m ² /s)	Q (kJ/mol)	$D^*(973.15 \text{ K})$
Perkins/ORNL[1]	1.30×10^{-5}	264.0104	8.77×10^{-20}
Mizouchi, SUS316[2]	1.13×10^{-7}	234	3.11×10^{-20}
Smith, Type 316[3]	6.30×10^{-6}	243	5.71×10^{-19}

Table 1: Paired Arrhenius parameters are retained within each study. All final-column values are in m²/s and are evaluated here.

Mizouchi used ⁵¹Cr penetration profiles and sputter microsectioning over 888–1173 K. Grain coarsening to 3–6 mm and stress relief helped isolate lattice transport. The SUS316 lattice fit is distinct from the paper’s 316L grain-boundary measurements.[2]

Smith’s value is a 50 K extrapolation below its 1023–1473 K measurement interval. The publisher abstract establishes that interval; the numerical fit is reproduced in Weck and Kim’s Table 2. The original Smith full article and regression uncertainties were not obtained. These reproductions are not independent measurements.[3, 4]

The fully inspected, non-extrapolated pair spans 3.11 – 8.77×10^{-20} m²/s. Including Smith expands the span to a factor of 18.3. The ORNL baseline lies inside both ranges.

Compare coefficients before combining them

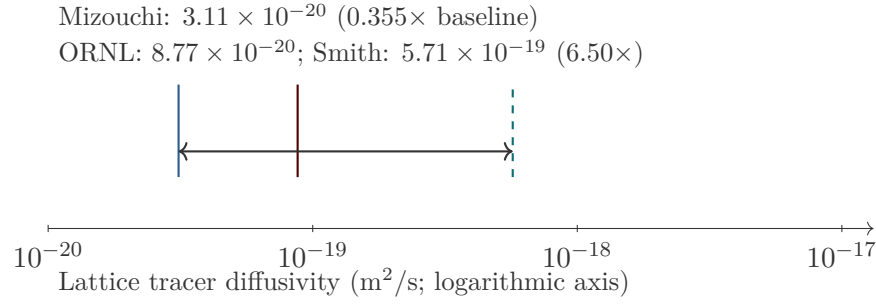


Figure 1: Experimental alternatives for sensitivity analysis. Line style identifies the extrapolated Smith value.

What atomistic simulations add

Weck and Kim used spin-polarized DFT/PBE ab initio molecular dynamics, 70–85 ps trajectories and mean-square-displacement slopes for a 316-like composition. Their bulk cell had 120 sites; removing four Cr atoms created 3.33% vacant sites and also changed the occupied-site composition. A 96-atom cell represented a $\Sigma 3$ twin.[4]

Simulated state	Reported at 750 °C	Fit evaluated at 700 °C
Pristine bulk	4.2×10^{-14}	2.78×10^{-14}
Four vacancies	3.8×10^{-11}	3.17×10^{-11}
$\Sigma 3$ twin cell	9.5×10^{-10}	9.16×10^{-10}

Table 2: Cr diffusivities in m^2/s . The last column is our 50 K extrapolation of rounded two-temperature fits, not a simulation reported at 700 °C.

The authors acknowledge a large discrepancy with experimental lattice diffusion. Our assessment is that the nearly stationary, fluctuating pristine MSD warrants resolved-jump and slope-uncertainty checks before using its absolute value. These simulations provide evidence of defect and pathway sensitivity; they do not justify increasing the upper experimental confidence bound by six orders of magnitude.

Bonny et al.’s EAM potential and atomistic kinetic Monte Carlo for Fe–10Ni–20Cr offer a route to longer vacancy-jump times. DFT defect calculations by Klaver et al. inform mechanisms in dilute austenitic Fe. Neither accessible record establishes a verified, concentrated-316 equilibrium coefficient at 700 °C for this assessment.[5, 6]

Microstructure and effective transport

A fast pathway is not a bulk coefficient

Mizouchi's grain-boundary fits give approximately $1.32\text{--}4.25 \times 10^{-10} \text{ m}^2/\text{s}$ when evaluated at 973.15 K for different metallurgical states. These are extrapolations of low-temperature type-C fits; higher-temperature type-B measurements support the solution-treated extrapolation under an assumed segregation-width product. Older fits imply substantially smaller values. Sampling different penetration tails and boundary populations contributes to the discrepancy.[2]

$$P = s\delta D_{\text{gb}}. \quad (2)$$

Equation meaning. P has units m^3/s ; s is a segregation factor and δ is the boundary width. Neither P nor D_{gb} can replace a volume-averaged alloy coefficient directly.

An effective transport model also needs boundary fraction, orientation, connectivity, exchange with grain interiors and the specimen's dislocation/precipitate state. Grain size alone does not provide these. We adopt no new homogenization formula here.

An actual corrosion experiment

Zheng inferred $D_{\text{eff}} = 4.2 \times 10^{-19} \text{ m}^2/\text{s}$ at 700 °C from 316 mass loss in FLiBe, in 316 capsules without graphite. Depletion profiles were compared with SEM/EDS, with grain-boundary attack and carbide formation discussed.[7] This is a valuable corrosion-level check, but includes the specimen microstructure and surface assumptions of the fitting model. It is neither an independent lattice-tracer measurement nor a FLiNaK calibration.

Screen assessed-mobility results carefully

A CALPHAD calculation for AISI316 gives $8.9 \times 10^{-19} \text{ m}^2/\text{s}$ at 750 °C, a different temperature. It inherits the experiments underlying its mobility assessment.[8] A separate DICTRA thesis lists $8.682 \times 10^{-16} \text{ m}^2/\text{s}$ at 700 °C, while its paired Arrhenius parameters give $7.96 \times 10^{-20} \text{ m}^2/\text{s}$. The approximately 1.09×10^4 disagreement between Tables 5.3 and 5.4 is unresolved, so those values are excluded from bounding.[9]

Use three evidence categories

Use experimental lattice fits as parameter alternatives; use atomistic calculations to assess mechanism and model discrepancy; use corrosion and fast-path evidence to test microstructure-dependent transport. Pooling all three into one numerical range would obscure what the model parameter means.

Propagating uncertainty into corrosion

Keep vacancy and tracer effects distinct

The retained model uses $S_v = x_{v,b}/x_{v,\text{thermal}}$, with $x_{v,\text{thermal}} = 2.91878 \times 10^{-8}$. Initial and far-boundary compositions coincide. Preserve occupied-site Fe:Ni:Cr ratios when changing S_v . For the frozen-coefficient analytical benchmark,

$$m_i = \frac{D_i^* x_{i,b} S_v}{RT}, \quad \mathbf{B} = \begin{pmatrix} m_{\text{Ni}} & 0 & -m_{\text{Ni}} \\ 0 & m_{\text{Cr}} & -m_{\text{Cr}} \\ -m_{\text{Ni}} & -m_{\text{Cr}} & m_{\text{Fe}} + m_{\text{Ni}} + m_{\text{Cr}} \end{pmatrix}, \quad \mathbf{D} = \mathbf{B}\mathbf{H}. \quad (3)$$

Equation meaning. $\mathbf{H}$ is the free-energy Hessian in $(x_{\text{Ni}}, x_{\text{Cr}}, x_v)$. Varying D_{Cr}^* changes all linked Cr terms, including B_{vv} , while Fe/Ni tracer inputs stay fixed. Vacancy-enhanced simulation values cannot be inserted as thermal inputs without checking for double counting.

The thermodynamic-only comparison retains $\mathbf{H}$ and only the diagonal of $\mathbf{B}$. Both options retain the same established thermodynamic model.

$$\mathbf{q}_s = \mathbf{q}_b - L\mathbf{D}^{-1}\mathbf{s}r_\infty, \quad r_{\text{limit}} = \frac{x_{\text{Cr},b}}{L(\mathbf{D}^{-1}\mathbf{s})_{\text{Cr}}}, \quad \mathbf{s} = (0, 1, -1)^T. \quad (4)$$

Here $r = j/(2FC_{\text{site}})$ is Cr-equivalent removal speed, not wall recession. Scaling all entries of $\mathbf{D}$ scales this ceiling; changing one tracer input need not do so proportionally.

Why all three interface conditions matter

Prescribed actual overpotential uses the retained linear surface law $r = hx_{\text{Cr},s}$, with h set by the chosen voltage. Mixed-potential benchmarks instead keep nonlinear activities and solve anodic–cathodic balance. Finite bath permits chemical depletion; maintained chemistry fixes the salt reservoir activities. The same analytical bulk diffusion kernels apply, with numerical boundary-history quadrature for mixed potential.

A finite bath approaches zero net current and finite accumulated removal. Its final alloy profile returns to the imposed far composition; terminal bath equilibrium depends on that composition and capacity, not diffusivity. Maintained chemistry and positive prescribed overpotential can sustain nonzero steady transfer from the far reservoir. Their cumulative removal has no finite long-time limit. Two years is an observation time, not necessarily steady state.

A defensible uncertainty workflow

1. Compare the three lattice fits at fixed S_v , then vary S_v at fixed tracer input; finally vary both. Treat the broad custom range as sensitivity, not a probability distribution.
2. Compare rates, accumulated removal and Ni/Cr/v profiles from 1 h to 17,520 h and the separate final asymptote. Keep length and interface parameters fixed for each comparison.
3. Obtain original fit uncertainties before claiming confidence intervals. Preserve paired D_0, Q and their covariance; separate microstructure and method discrepancy.

Sources and companion analysis

The adjacent evidence CSV retains numerical parameters, source locations and qualifications. The companion explorer provides independent controls for S_v and D_{Cr}^* across all three interface conditions. These are frozen-coefficient analytical/semi-analytical benchmarks, not new full nonlinear MOOSE simulations. Interface scales and bath capacity retain their earlier diagnostic values; the literature envelope is not an experimental calibration of those quantities.

$$\text{Var}(\ln D) = \text{Var}(\ln D_0) + \frac{\text{Var}(Q)}{(RT)^2} - \frac{2 \text{Cov}(\ln D_0, Q)}{RT}. \quad (5)$$

References

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